

Fundamentals of Financial Markets
2023.09.05.
Solution

Name:

Neptun code:

Mark one answer clearly. No explanation is necessary. Each correct answer is worth 2 points, each incorrect answer is -1 point! The minimum requirement 40%, that is 16 points.

	1	2	3	4	5	6	7	8	9	10
A		X		X					X	
B	X		X			X	X			
C								X		
D					X					X

	11	12	13	14	15	16	17	18	19	20
A	X		X			X				X
B		X								
C					X				X	
D				X			X	X		

1. Suppose that discount factors for different maturities (one, two and three years) are given as 96%, 90% and 82%, respectively. What is the yield (annual compounding) on a 2-year zerocoupon Treasury bond?

- a) 5.27%
- b) 5.4%**
- c) 6.45%
- d) 11.11%

$$r_2 = (1/0.9)^{(1/2)} - 1 = 5.4\%$$

2. A bank quotes you an interest rate of 8% per annum with quarterly compounding. What is the equivalent rate with continuous compounding?

- a) 7.92%**
- b) 8%
- c) 8.24%
- d) 32%

$$y = \ln((1+8\%/4)^4) = 7.92\%$$

3. The yield on 1-year zero-coupon bonds is currently 4%, and on 2-year zeros it is 7% (per annum with continuous compounding). The Treasury plans to issue a 2-year maturity coupon bond, paying coupons once per year with a coupon rate of 10%. The face value of the bond is \$100. At what price will the bond sell?

- a) \$104.95
- b) \$105.24**
- c) \$105.69
- d) \$111.15

1	10	4%	9,608
2	110	7%	95,629
			105,237

4. Suppose that today the 3-months LIBOR is 7%. The Treasury issues a 2-year floating-rate bond, paying coupons four times per year with a coupon rate of LIBOR. The face value of the bond is \$100. If at the end of the first month the yield curve flattens out at 5% (annual compounding), what will be the price of the floating-rate bond?

- a) \$100.93**
- b) \$101.34
- c) \$106.13
- d) \$106.57

$$P = 101,75/1,05^{(2/12)} = \$100.93$$

5. Consider a 10-year bond pays an annual coupon of \$100 and sells at par value of \$1,000 at the beginning of the year. Suppose the yield falls and the price increases to \$1,050 at the end of the year (after coupon payment). What will be the 1 year holding period return?

- a) 0%
- b) 5%
- c) 10%
- d) 15%**

$$r = (1050+100)/1000-1 = 15\%$$

6. Suppose that the 1-year, 2-year and 3-year zero rates are 5%, 6% and 7%, respectively (continuous compounding). What is the 2-year par yield?

- a) 5.97%
- b) 6.15%**
- c) 6.91%
- d) 7.15%

1	5%	0,9512	0,0615
2	6%	0,8869	
		1,8381	

7. The current yield curve for zero-coupon bonds is as follows 9%, 10% and 11% (one two and three years with annual compounding). What is the implied ${}_1f_2$ forward rate with continuous compounding?

- a) 9.44%
- b) 10.44%
- c) 11.44%
- d) 12.44%

1	9%	0,9174	0,9008	0,1044
2	10%	0,8264		
3	11%	0,7312		

8. Assuming the pure expectations theory is correct, an downward-sloping yield curve implies:

- a) Interest rates are expected to increase in the future.
- b) Longer-term bonds are riskier than short-term bonds.
- c) Interest rates are expected to decline in the future.
- d) Short-term bonds are riskier than longer-term bonds.

9. Suppose that discount factors for different maturities (one and two years) are given as 92%, and 84%, respectively. An investor purchased a 5-year zero-coupon bonds three years ago for \$800. The face value of the bond is \$1000. What will be the holding period return (per annum, annually compounding) to the investor who sells the bond 1 year from today if the yield curve will be the same as today?

- a) 3.56%
- b) 5.56%
- c) 7.56%
- d) 9.56%

$$HPR = (920/800)^{0,25-1} = 3,56\%$$

10. The yield to maturity on 1-year zero-coupon bonds is currently 8% (annual compounding); the YTM on 2-year zeros is 10% (continuous compounding). The Treasury plans to issue a 2-year maturity coupon bond, paying coupons once per year with a coupon rate of 12%. The face value of the bond is \$100. What is the market expectation of the price that the bond will sell for next year (after coupon payment), if the yield curve will be the same as today?

- a) \$99.39
- b) \$101.39
- c) \$102.39
- d) \$103.39

1	12	8%	0,9231		
2	112	10%	0,8187	112	103,389

11. Which of the following is TRUE according to the carry trader?

- a) The carry trader believes that the term structure will not change during the investment period.
- b) The carry trader believes that the term structure will change during the investment period based on the liquidity preference hypothesis.
- c) The carry trader believes in the expectations hypothesis.
- d) If the yield curve is downward sloping the carry trader sells short-term bonds and invests in long-term bonds

12. Which statement is FALSE?

- a) funds in the bank's reserve account are called federal funds
- b) the fed funds rate is the rate of interest on long-term loans among financial institutions.
- c) The London Interbank Offered Rate (LIBOR) is the rate at which large banks in London are willing to lend money among themselves.
- d) LIBOR interest rates may be tied to currencies other than the U.S. dollar.

13. Which of the following correctly describes a repurchase agreement?

- a) The sale of a security with a commitment to repurchase the same security at a specified future date and a designated price.
- b) The sale of a security with a commitment to repurchase the same security at a future date left unspecified, at a designated price.
- c) The purchase of a security with a commitment to purchase more of the same security at a specified future date.
- d) The sale of a security with a commitment to repurchase the same security at a specified future date and a discounted price.

14. You predict that interest rates are about to fall. Which bond will give you the highest capital gain?

- a) Short position, High coupon, 2-years maturity.
- b) Short position, Low coupon, 10-year maturity.
- c) Long position, High coupon, 2-years maturity.
- d) Long position, Low coupon, 10-year maturity.

15. Find the duration of a 8% coupon bond making annual coupon payments (face value of the bond is \$100) if it has 3 years until maturity and the yield curve is flat at 6% (continuous compounding).

- a) 2.59 years
- b) 2.69 years
- c) 2.79 years
- d) 2.89 years

1	8,00	0,06	7,53	0,07	0,07
2	8,00	0,06	7,10	0,07	0,14
3	108	0,06	90,21	0,86	2,58
			104,84	1,00	2,79

16. An investor enters into a long forward contract to buy 100.000 British pounds for U.S dollars at an exchange rate of 1.3000 USD per pound. How much does the investor gain or lose (in dollars) if the exchange rate at the end of the contract is 1.2700?

- a) Lose 3000 USD
- b) Gain 3000 USD
- c) Gain 2307,7 USD
- d) Lose 2307,7 USD

17. When does margin call happen, and how much must the investor pay? If the balance in the margin account falls below the ...

- a) maintenance margin, he must pay up to the maintenance margin level.
- b) initial margin, he must pay up to the maintenance margin level.
- c) the initial margin, he must pay up to the initial margin level.
- d) maintenance margin, he must pay up to the initial margin level.

18. An investor sells a one-year maturity European call on a share for \$10. The stock price is \$50 and the strike price is \$52. Under what circumstances will the option be exercised and the investor makes a loss? You can borrow or lend money for 1 year at 0%.

- a) If the spot price at the maturity is under \$42
- b) If the spot price at the maturity is between \$42 and \$52
- c) If the spot price at the maturity is between \$52 and \$62
- d) If the spot price at the maturity is over \$62

19. An investor buys a one-year maturity European call on a share for \$10. The stock price is \$50 and the strike price is \$52. Under what circumstances will the option be exercised and the investor makes a loss? You can borrow or lend money for 1 year at 0%.

- a) If the spot price at the maturity is under \$42
- b) If the spot price at the maturity is between \$42 and \$52
- c) If the spot price at the maturity is between \$52 and \$62
- d) If the spot price at the maturity is over \$62

20. An investor sells a one-year maturity European put on a share for \$10. The stock price is \$50 and the strike price is \$52. Under what circumstances will the option be exercised and the investor makes a loss? You can borrow or lend money for 1 year at 0%.

- a) If the spot price at the maturity is under \$42
- b) If the spot price at the maturity is between \$42 and \$52
- c) If the spot price at the maturity is between \$52 and \$62
- d) If the spot price at the maturity is over \$62